Post-Processing Guarantees for Classification under Linear-Fractional Performance Metrics

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Abstract

This paper studies binary classification under generalized performance metrics, going beyond standard misclassification error. We focus on linear-fractional measures—including the F-score and Jaccard index—which are widely used in class-imbalanced settings. We first provide a unified characterization of the optimal classifier, showing that it admits a thresholding structure on the regression function. This reduces the original infinite-dimensional, non-decomposable optimization problem to the estimation of a single scalar parameter. Building on this result, we analyze plug-in classifiers that combine a regression estimator with a data-driven estimate of the optimal threshold. We show that this threshold is characterized as the solution of a fixed-point equation depending only on the marginal distribution, enabling its estimation using unlabeled data and avoiding standard validation-based procedures. Our main contribution is a finite-sample analysis of this approach. We derive excess risk bounds that decompose the error into contributions from regression estimation, threshold estimation, and sampling effects, providing a modular understanding of learning under non-decomposable metrics. Under standard margin assumptions, we further establish fast convergence rates. Our results yield a unified theoretical framework for plug-in classification under linear-fractional metrics, extending prior analyses beyond specific measures such as the F-score. Empirical results on synthetic and real datasets support our theoretical findings.

1 Introduction

Binary classification is a fundamental problem in machine learning, where the goal is to predict binary outcomes from input features. Classifier performance is traditionally evaluated using classification accuracy, i.e., the expected proportion of correct predictions. However, accuracy can be inadequate in many practical scenarios such as medical diagnosis, fraud detection, click-through rate prediction, and information retrieval, particularly under class imbalance or label noise (Lewis, 1995; Drummond & Holte, 2005; Gu et al., 2009; He & Garcia, 2009; Menon et al., 2015).

To address these limitations, alternative performance metrics have been widely adopted, including the F_β-measure, Jaccard index, and balanced error rate (Joachims, 2005; Brodersen et al., 2010; Ye et al., 2012; Berman et al., 2018; Charoenphakdee et al., 2019). These metrics better capture trade-offs between false positives and false negatives and can be expressed as functions of the confusion matrix entries (Dembczyński et al., 2017), but are typically non-decomposable, meaning that they cannot be expressed as expectations of pointwise losses. As a consequence, standard empirical risk minimization techniques (Geer, 2000) are not directly applicable, making both optimization and statistical analysis significantly more challenging.

A prominent approach to this problem considers the class of linear-fractional performance metrics, which express performance as ratios of linear functions of the confusion matrix entries (Koyejo et al.,

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2014). This family encompasses many commonly used criteria, including the F_β-measure, Jaccard index, and weighted accuracy. A key insight is that, despite their non-decomposability, the optimal classifier for such metrics can be characterized as a thresholding rule on the regression function $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$, with a metric-dependent threshold (Koyejo et al., 2014).

Building on this structure, a natural strategy is to adopt a plug-in approach: first estimate the regression function, then determine an optimal threshold. However, the threshold depends on the underlying data distribution and is typically unknown, making its estimation a central challenge. Moreover, existing approaches often rely on labeled validation data or computationally intensive procedures.

In this work, we develop a general framework for classification under linear-fractional metrics from a post-processing perspective. Starting from an arbitrary estimator of the regression function, we characterize the optimal threshold as the solution of a fixed-point equation and propose to estimate it using only unlabeled data. This leads to a simple and efficient procedure, particularly well suited to settings where labeled data is scarce but unlabeled data is abundant.

Related work. The study of classification under non-decomposable metrics has attracted increasing attention in recent years. A foundational contribution is due to Koyejo et al. (2014), who introduced the class of linear-fractional performance metrics and characterized the corresponding optimal classifiers. Building on this result, two main approaches have emerged. Plug-in methods estimate the regression function and then tune the threshold (Koyejo et al., 2014; Narasimhan et al., 2014, 2015; Yan et al., 2018). Alternatively, cost-sensitive reductions iteratively solve weighted empirical risk minimization problems (Koyejo et al., 2014; Parambath et al., 2014).

Another line of work focuses on surrogate-based approaches, where the original objective is replaced by a tractable surrogate utility. For instance, Bao et al. (2020) propose calibrated surrogate objectives for linear-fractional metrics and study conditions under which surrogate maximization is consistent with respect to the target metric. While these methods enable efficient optimization, their statistical analysis typically focuses on asymptotic consistency rather than finite-sample guarantees.

Closer to our work, post-processing approaches have been studied for specific metrics such as the F-score. In particular, Chzhen (2020) propose a post-processing procedure that calibrates any score-based classifier via thresholding, using unlabeled data and achieving minimax optimal rates under standard nonparametric assumptions. However, these results are tailored to specific metrics and do not directly extend to the broader class of linear-fractional utilities.

Contributions. Our work builds on and generalizes some of the above ideas. We establish finite-sample post-processing guarantees for plug-in classifiers obtained by thresholding an estimate of the regression function, where the threshold is defined as the solution of a fixed-point equation. We provide a new and simple derivation of the optimal threshold classifier based on this fixed-point characterization, recovering the result of Koyejo et al. (2014), and prove the optimality of the resulting decision rule among all measurable classifiers. We introduce a post-processing procedure that estimates the optimal threshold using only unlabeled data. Finally, we derive finite-sample excess utility bounds that explicitly decompose the contributions of regression error, threshold estimation error, and sampling error, and that adapt under margin assumptions. These results extend existing analyses beyond specific metrics such as the F-score (Chzhen, 2020) and provide a modular perspective on learning under non-decomposable evaluation criteria.

Limitations. First, the overall performance of the method depends on the quality of the regression estimator, whose computation can be costly in high-dimensional or large-scale settings. Second, our theoretical analysis is limited to the standard binary classification setting and multi-class or structured prediction problems are left for future work. In addition, the thresholding structure of optimal classifiers for linear-fractional metrics has already been studied in Koyejo et al. (2014); our contribution lies in providing an alternative probabilistic derivation and, in particular, a new original finite-sample analysis. Finally, for the F_1-score metric, post-processing guarantees were previously obtained in Chzhen (2020); our results recover them as a special case within a more general framework.

Organization. The paper is organized as follows. In Section 2 we introduce the mathematical setting and the class of linear-fractional performance metrics. In Section 3 we derive the optimal threshold via a fixed-point characterization, recovering the form in Koyejo et al. (2014), and establish

the optimality of the resulting classifier. In Section 4 we present our main results, namely finite-sample post-processing excess utility bounds, both without assumptions and under margin conditions. Finally, in Section 5 we present numerical simulations and real data experiments.

2 Setting

Let us consider the classical supervised learning setting. Let (X, Y) be a pair of random variables taking values in $\mathbb{R}^d \times \{0, 1\}$ with joint distribution P. We denote by P_X the marginal distribution of $X \in \mathbb{R}^d$, by $p := \mathbb{P}(Y = 1)$ the class prior, and by $\eta(X) := \mathbb{E}[Y \mid X]$ the regression function. Finally, let $\mathcal{G}$ be the set of all measurable classifiers $g : \mathbb{R}^d \to \{0, 1\}$.

Given a classifier g, there are several ways to evaluate its predictive performance. The most standard choice is the misclassification error $P(Y \neq g(X))$. However, in many applications practitioners are interested in alternative metrics that capture different trade-offs, for instance between precision and recall. In this work, we focus on a class of utility measures that is general enough to recover popular metrics such as accuracy, F-score, Jaccard index, and others as special cases. To this end, we introduce the class of linear-fractional performance measures. Given coefficients $\boldsymbol{c} = (c_0, c_1, c_2) \in \mathbb{R}^3$ and $\boldsymbol{d} = (d_0, d_1, d_2) \in \mathbb{R}^3$, the performance of a classifier $g \in \mathcal{G}$ is defined as

$$\mathrm{U}_{(c,d)}(g) := \frac{c_0 + c_1 \mathbb{P}[Y = 1, g(X) = 1] + c_2 \mathbb{P}[g(X) = 1]}{d_0 + d_1 \mathbb{P}[Y = 1, g(X) = 1] + d_2 \mathbb{P}[g(X) = 1]}. \tag{1}$$

This formulation encompasses a wide range of commonly used metrics, including accuracy, F-score, and the Jaccard index, for appropriate choices of $\boldsymbol{c}$ and $\boldsymbol{d}$. Our goal is to characterize the optimal classifier $g^* \in \mathcal{G}$ maximizing this utility, i.e.,

$$g^* = \arg\max_{g \in \mathcal{G}} \mathrm{U}_{(c,d)}(g). \tag{2}$$

This optimization problem is non-convex and, in general, difficult to solve. In the next section, we show how its structure can be significantly simplified.

Metric	Expression	(c_0, c_1, c_2)	(d_0, d_1, d_2)
Accuracy	$\mathbb{P}(Y = g(X))$	$(1 - p,\ 2,\ -1)$	$(1,\ 0,\ 0)$
$\mathbf{F}_b$-score	$\dfrac{(1 + b^2)\mathbb{P}(Y = 1, g(X) = 1)}{b^2 p + \mathbb{P}(g(X) = 1)}$	$(0,\ 1 + b^2,\ 0)$	$(b^2 p,\ 0,\ 1)$
Jaccard	$\dfrac{\mathbb{P}(Y = 1, g(X) = 1)}{\mathbb{P}(Y = 1, g(X) = 0) + \mathbb{P}(g(X) = 1)}$	$(0,\ 1,\ 0)$	$(p,\ -1,\ 1)$
AM	$\dfrac{1}{2} \displaystyle\sum_{i \in \{0,1\}} \mathbb{P}(g(X) = i \mid Y = i)$	$\left(\frac{1}{2}, \frac{1}{2p} + \frac{1}{2(1-p)}, -\frac{1}{2(1-p)}\right)$	$(1,\ 0,\ 0)$
Recall	$\mathbb{P}(g(X) = 1 \mid Y = 1)$	$(0,\ 1,\ 0)$	$(p,\ 0,\ 0)$

Table 1: Choices of parameters $\boldsymbol{c}$ and $\boldsymbol{d}$ for common performance metrics.

3 Optimality of thresholding functions

In this section, we show that simple thresholding rules are optimal. We begin by restricting the optimization problem in Eq. (2) to classifiers of the form $g_\theta(X) = \mathbb{1}\{\eta(X) \geqslant \theta\}$. We denote this parametric class by $\mathcal{G}_\theta = \{g_\theta : g_\theta(\cdot) = \mathbb{1}\{\eta(\cdot) \geqslant \theta\}\}$ and aim to solve

$$g_{\theta^*}(X) \in \arg\max_{\mathcal{G}_\theta} \mathrm{U}(g). \tag{3}$$

Before proceeding, we introduce assumptions on the regularity of the distribution and on the coefficients $\boldsymbol{c}$ and $\boldsymbol{d}$ in Eq. (1), which will be used throughout the paper.

Assumption 1. *Let $Z = \eta(X)$ and denote by F_Z its cumulative distribution function. Assume that F_Z is continuously differentiable on $(0, 1)$ with density f_Z, and that f_Z is bounded on $(0, 1)$.*

The above assumption is common in the literature, see e.g. Assumption 3.1 in Gaucher et al. (2023) or Assumption 2 in Yan et al. (2018).

Assumption 2. *The coefficients $c_0, c_1, c_2, d_0, d_1, d_2$ in Eq. (1) satisfy:*

i)
$$c_1 \geqslant 0,$$

ii)
$$d_0 + \min\{\min\{d_1, 0\} + d_2, 0\} > 0, \qquad d_2 \geqslant 0$$

iii)
$$\begin{cases} d_2 c_1 > c_2 d_1 \\ \frac{c_0 d_2 - d_0 c_2}{c_2 d_1 - d_2 c_1} \leqslant p \\ d_0 c_1 - c_0 d_1 > (c_0 d_2 - d_0 c_2)_+ \end{cases}$$

Assumption 2 is mild and satisfied by standard performance metrics such as accuracy, F-score, and the Jaccard index. Condition (i) ensures that the numerator is increasing in the true positives, which is consistent with common metrics. Condition (ii) guarantees that the utility is well-defined by preventing division by zero or negative denominators. Condition (iii) is a technical requirement ensuring uniqueness of the optimal threshold θ^* by ruling out pathological cases such as flat regions or multiple optima.

We have the following result characterizing the optimal threshold within $\mathcal{G}_\theta$.

Lemma 1. *Consider the utility function in Eq. (1) under conditions in Assumption 1 and 2. Consider only classifiers $g \in \mathcal{G}_\theta$ then:*

$$\mathbb{E}[(\eta(X) - \theta^*)_+] = \frac{c_0 d_1 - c_1 d_0}{c_2 d_1 - c_1 d_2}\theta^* + \frac{c_0 d_2 - c_2 d_0}{c_2 d_1 - c_1 d_2} \tag{4}$$

where θ^* is the optimal parameter such that $g_{\theta^*}(X) = \mathbb{1}\{\eta(X) \geqslant \theta^*\}$ maximizes (1) in $\mathcal{G}_\theta$.

We refer to Appendix A for the proof. The lemma above characterizes the optimal threshold as the solution of a fixed-point equation that depends only on the marginal distribution P_X. This is particularly appealing in applications where labeled data are scarce or expensive, while unlabeled samples are abundant. We discuss how to solve Eq. (4) in practice in Section 4.

While the threshold structure of optimal classifiers for linear-fractional metrics is already introduced in Koyejo et al. (2014), our derivation relies on a direct probabilistic argument rather than functional-analytic tools such as Fréchet derivatives, which typically require additional technical care.

Remark 1. *In case $d_2 c_1 = c_2 d_1$ (e.g., the AM measure; see Table 1), condition (iii) in Assumption 2 is replaced by*

$$\begin{cases} d_2 c_1 = c_2 d_1 \\ c_1 d_0 > d_1 c_0 \\ \frac{d_0 c_2 - c_0 d_2}{c_0 d_1 - d_0 c_1} \in [0, 1] \end{cases}$$

and the optimal threshold is given explicitly by $\theta^* = \frac{d_0 c_2 - c_0 d_2}{c_0 d_1 - d_0 c_1}$.

Given Lemma 1, we now show that the classifier $g_{\theta^*}(x) = \mathbb{1}\{\eta(x) \geqslant \theta^*\}$ is optimal over the entire class of measurable classifiers $\mathcal{G}$.

Lemma 2. *Let $g \in \mathcal{G}$ be any classifier, then:*

$$\mathrm{U}_{(c,d)}(g_{\theta^*}) - \mathrm{U}_{(c,d)}(g) = \frac{c_1 \mathbb{E}[|\eta(X) - \theta^*| \mathbb{1}\{g_{\theta^*}(X) \neq g(X)\}]}{d_0 + d_1 \mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + d_2 \mathbb{P}[g(X) = 1]} \geqslant 0, \tag{5}$$

which shows that g_{θ^*} is optimal among all measurable classifiers.

The proof is given in Appendix A. The result shows that optimal classifiers for the utility in Eq. (1) can be restricted to thresholding rules of the form $g_\theta(x) = \mathbb{1}\{\eta(x) \geqslant \theta\}$. This is particularly useful, as the original optimization problem over all measurable classifiers—an infinite-dimensional problem with a non-linear fractional objective—reduces to the one-dimensional problem of estimating a scalar threshold $\theta^* \in [0, 1]$.

For completeness, we conclude with another useful identity.

Lemma 3 (Lemma 4.1 in Gaucher et al. (2023))**.** *The utility function given in Eq. (1) and evaluated for g_{θ^*} takes the form*

$$\mathrm{U}_{(c,d)}\left(g_{\theta^*}\right) = \frac{c_2 + \theta^* c_1}{d_2 + \theta^* d_1}.$$

4 Post-processing Bounds

In this section, we derive post-processing bounds for plug-in estimators without imposing additional assumptions on the joint distribution P of (X, Y). The key idea is to decouple the estimation of the regression function from the estimation of the optimal threshold.

We assume access to two independent datasets: a labeled dataset $\mathcal{D}_n^{\mathrm{L}} = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ consisting of n i.i.d. samples from P, and an unlabeled dataset $\mathcal{D}_N^{\mathrm{U}} = \{X_{n+1}, \ldots, X_{n+N}\}$ consisting of N i.i.d. samples from the marginal distribution P_X, with $N \geqslant n$. In many practical scenarios, unlabeled data are significantly easier and cheaper to collect than labeled samples. This setting is therefore particularly appealing, as it allows one to leverage large amounts of unlabeled data to estimate the optimal threshold accurately, while using labeled data only to learn the regression function. We denote expectations with respect to $\mathcal{D}_N^{\mathrm{U}} \sim P_X^{\otimes N}$ by $\mathbb{E}_{\mathcal{D}_N^{\mathrm{U}}}$, while $\mathbb{E}$ denotes a generic expectation over both labelled and unlabelled data.

Let $\hat{\eta}$ be an estimator of the regression function η, constructed from the labeled dataset $\mathcal{D}_n^{\mathrm{L}}$. Using $\hat{\eta}$ together with the unlabeled data $\mathcal{D}_N^{\mathrm{U}}$, we estimate the optimal threshold θ^* by solving the empirical counterpart of Eq. (4). In practice, this can be done, for instance, via a bisection method applied to

$$\hat{R}(\theta) = \frac{c_0 d_1 - c_1 d_0}{c_2 d_1 - c_1 d_2}\theta + \frac{c_0 d_2 - c_2 d_0}{c_2 d_1 - c_1 d_2} - \hat{\mathbb{E}}_N(\hat{\eta}(X) - \theta)_+, \tag{6}$$

where $\hat{\mathbb{E}}_N$ denotes the empirical expectation with respect to the measure $\frac{1}{N}\sum_{i=n+1}^{n+N} \delta_{X_i}$ evaluated on the unlabelled data.

To guarantee the existence of a root of $\hat{R}$, we impose an empirical analogue of condition (iii) in Assumption 2:

$$\frac{c_0 d_2 - d_0 c_2}{c_2 d_1 - d_2 c_1} \leqslant \hat{\mathbb{E}}_N(\hat{\eta}(X)), \qquad \text{a.s.} \tag{7}$$

which ensures $\hat{R}(0) \leqslant 0$, and with $\hat{R}(1) \geqslant 0$ given by Assumption 2 (iii).

Remark 2. *Condition (7) is not restrictive in practice, for two reasons. First, it holds automatically for the metrics in Table 1. For the F_b-score and the Jaccard index, its left-hand side is zero, while the right-hand side is strictly positive since $\hat{\eta} \in (0, 1]$. For accuracy, the AM measure and recall, $c_1 d_2 = c_2 d_1$, so we are in the setting of Remark 1: the threshold is explicit and (7) plays no role. More generally, the condition holds surely whenever its left-hand side is non-positive. When it is positive, requiring it almost surely amounts to $\hat{\eta}(x)$ being at least that constant for $\mathbb{P}_X$-almost every x, i.e. a condition on the estimator rather than on the unlabeled sample. Second, a violation is algorithmically benign. Under Assumption 2 (iii), $\hat{R}$ is non-decreasing and $\hat{R}(1) \geqslant 0$ regardless of the data. If Eq. (7) fails, then $\hat{R} > 0$ on all of $[0, 1]$, and our implementation returns $\hat{\theta} = 0$, i.e. it classifies every point as positive. This is the meaningful answer rather than a failure mode: by the first-order computation in the proof of Lemma 1, a positive criterion means that the utility is decreasing in θ, so it is maximized at $\theta = 0$.*

We define the resulting plug-in classifier as

$$\hat{g}(\cdot) := \mathbb{1}\{\hat{\eta}(\cdot) \geqslant \hat{\theta}\}. \tag{8}$$

The goal of this section is to control the excess utility of $\hat{g}$:

$$\mathrm{U}(g_{\theta^*}) - \mathrm{U}(\hat{g}) = \frac{c_1 \mathbb{E}[|\eta(X) - \theta^*| \mathbb{1}\{g_{\theta^*}(X) \neq \hat{g}(X)\}]}{d_0 + d_1 \mathbb{E}[\eta(X)\mathbb{1}\{\hat{g}(X) = 1\}] + d_2 \mathbb{P}[\hat{g}(X) = 1]}, \tag{9}$$

where we write U instead of $\mathrm{U}_{(c,d)}$ for simplicity.

4.1 Post-processing bound under minimal assumptions

We now present our main post-processing bound for the plug-in classifier $\hat{g}$. The result provides a finite-sample control of the excess utility without imposing additional assumptions on the distribution P. In particular, it shows how the performance of $\hat{g}$ depends on three key components: the accuracy of the regression estimator $\hat{\eta}$, the amount of unlabeled data used to estimate the threshold, and the precision of the numerical procedure used to compute $\hat{\theta}$.

Proposition 1. *Consider the utility function in Eq. (1) under Assumptions 1 and 2, together with condition (7). Let $\hat{\eta}$ be an estimator of η such that $\hat{\eta}(x) \in (0,1]$ almost surely, and define $\hat{g}(x) = \mathbb{1}\{\hat{\eta}(x) \geqslant \hat{\theta}\}$, where $\hat{\theta}$ is obtained by applying the bisection algorithm to Eq. (6) with $K_{\max} \in \mathbb{N}$ iterations. Then,*

$$\mathrm{U}(g_{\theta^*}) - \mathbb{E}\mathrm{U}(\hat{g}) \leqslant c_1 K_0 (1 + K_1) \mathbb{E}\|\eta - \hat{\eta}\|_1 + c_1 K_0 K_1 \left(\sqrt{\frac{\pi p}{N}} + \frac{4}{N} \right) + c_1 K_0 2^{-K_{\max}} \quad (10)$$

with $K_0 = 1/(d_0 + \min\{d_1 + d_2, 0\} p)$ and $K_1 = \frac{c_2 d_1 - c_1 d_2}{c_0 d_1 - c_1 d_0}$.

The proof is given in Appendix A. The bound decomposes the excess utility into three contributions, corresponding to distinct sources of error in the plug-in procedure.

The first term, proportional to $\mathbb{E}\|\eta - \hat{\eta}\|_1$, captures the error due to estimating the regression function. This is the standard term in plug-in analyses and reflects the intrinsic difficulty of learning η from labeled data. Notably, it is independent of the amount of unlabeled data.
The second term depends on the size N of the unlabeled dataset and quantifies the error arising from estimating the optimal threshold θ^* through its empirical counterpart. This highlights a key advantage of the approach: since unlabeled data are often cheaper and more abundant, the threshold estimation error can be reduced independently by increasing N. The rate $O(N^{-1/2})$ reflects concentration of the empirical estimate around its population counterpart.
The third term corresponds to the optimization error induced by the numerical procedure used to compute $\hat{\theta}$. In our case, this is controlled by the number of iterations $K_{\max}$ of the bisection algorithm and decays exponentially fast. This shows that the optimization step is not a bottleneck, as a moderate number of iterations suffices to make this term negligible.

Overall, the bound provides a modular view of the plug-in approach: regression estimation, threshold estimation, and numerical optimization contribute additively to the final performance. This decomposition clarifies how each component affects the excess utility and shows that access to large amounts of unlabeled data can substantially improve performance without requiring additional labeled samples.

Remark 3 (F-score)**.** *For the F_b-score, $c_0 = 0$, $c_1 = 1 + b^2$, $c_2 = 0$, $d_0 = b^2 p$, $d_1 = 0$, $d_2 = 1$, so that $K_0 = K_1 = \frac{1}{b^2 p}$. Proposition 1 gives*

$$F_b(g_{\theta^*}) - \mathbb{E}F_b(\hat{g}) \leq \frac{1 + b^2}{b^2 p} \left[\left(1 + \frac{1}{b^2 p}\right) \mathbb{E}\|\eta - \hat{\eta}\|_1 + \frac{1}{b^2 p} \left(\sqrt{\frac{\pi p}{N}} + \frac{4}{N} \right) + 2^{-K_{\max}} \right].$$

Moreover, by Lemma 3, $F_b(g_{\theta^*}) = (1 + b^2)\theta^*$. Hence the bound on $\mathbb{E}|\theta^* - \hat{\theta}|$ in Eq. (19) yields

$$\mathbb{E}\left|F_b(g_{\theta^*}) - (1 + b^2)\hat{\theta}\right| \leq (1 + b^2) \left(\frac{1}{b^2 p} \left(\mathbb{E}\|\eta - \hat{\eta}\|_1 + \sqrt{\frac{\pi p}{N}} + \frac{4}{N} \right) + 2^{-K_{\max}} \right),$$

which is exactly Proposition 3 in Chzhen (2020).

4.2 Post-processing bound under margin assumption

We now refine the previous analysis under additional assumptions on the distribution P, which allow us to obtain improved convergence rates. The key idea is that when the regression function $\eta(X)$ does not concentrate too heavily around the optimal threshold θ^*, classification errors become less likely, leading to faster decay of the excess utility.

We introduce the following margin condition, which quantifies how much probability mass is concentrated near the decision boundary.

Assumption 3 (Margin Assumption)**.** *The distribution P of $(X, Y) \in \mathbb{R}^d \times \{0, 1\}$ satisfies the α-margin condition if there exist constants $M > 0$ and $\alpha \geqslant 0$ such that*

$$\mathbb{P}(0 < |\eta(X) - \theta^*| \leqslant t) \leqslant M t^{\alpha} \quad \forall t > 0. \quad (11)$$

This condition ensures that the probability mass near the threshold θ^* decays polynomially. As a consequence, small errors in estimating η or θ^* translate into a smaller probability of misclassification, yielding faster convergence rates.

We are now ready to state an improved version of Proposition 1.

Proposition 2. *Consider the utility function in Eq. (1) under Assumptions 1 and 2, together with condition (7) and the margin assumption 3 with $\alpha > 0$, and let $q \in (1, \infty)$. let $\hat{\eta}$ be an estimator of η such that $\hat{\eta}(x) \in (0, 1]$ almost surely, and define $\hat{g}(x) = \mathbb{1}\{\hat{\eta}(x) \geqslant \hat{\theta}\}$, where $\hat{\theta}$ is obtained via the bisection algorithm applied to Eq. (6) with $K_{\max} \in \mathbb{N}$ iterations. Then*

$$\mathrm{U}(g_{\theta^*}) - \mathbb{E}\mathrm{U}(\hat{g}) \leqslant C_0(q, \alpha, M, P, \boldsymbol{c}, \boldsymbol{d})\mathbb{E}\|\eta - \hat{\eta}\|_q^{\frac{q(1+\alpha)}{q+\alpha}} + \frac{C_1(\alpha, M, P, \boldsymbol{c}, \boldsymbol{d})}{\sqrt{N^{1+\alpha}}} + \frac{C_2(\alpha, M, P, \boldsymbol{c}, \boldsymbol{d})}{N^{1+\alpha}} + \\ + C_3(\alpha, M, P, \boldsymbol{c}, \boldsymbol{d})\mathbb{E}\|\eta - \hat{\eta}\|_1^{1+\alpha} + \frac{C_4(\alpha, M, P, \boldsymbol{c}, \boldsymbol{d})}{2^{(1+\alpha)K_{max}}} \tag{12}$$

where $C_0 = 2^{\frac{q(1+\alpha)}{q+\alpha}} c_1 K_0 \frac{q+\alpha}{q} \left(\frac{q}{\alpha}\right)^{\frac{\alpha}{q+\alpha}} M^{\frac{q-1}{q+\alpha}}$, $C_1 = (1+\alpha)\,\Gamma\!\left(\frac{1+\alpha}{2}\right) c_1 K_0 M K_1^{1+\alpha}\left(2^{2+4\alpha}\sqrt{p} + 2^{\frac{1+7\alpha}{2}}\right)$, $C_2 = 2^{4+5\alpha}(1+\alpha)\,\Gamma(1+\alpha)\, c_1 K_0 M K_1^{1+\alpha}$, $C_3 = 2^{1+4\alpha} c_1 K_0 M K_1^{1+\alpha}$, $C_4 = c_1 K_0 M 2^{1+2\alpha}$, $K_0 = 1/(d_0 + \min\{d_1 + d_2, 0\}p)$, $K_1 = (c_2 d_1 - c_1 d_2)/(c_0 d_1 - c_1 d_0)$.

The proof is provided in Appendix A. Compared to the assumption-free bound, the margin condition yields improved convergence rates. In particular, the dependence on the regression error $\|\eta - \hat{\eta}\|_q$ now involves an exponent strictly larger than one when $\alpha > 0$, reflecting reduced sensitivity to estimation errors away from the decision boundary. Similarly, the dependence on the unlabeled sample size improves to rates of order $N^{-(1+\alpha)/2}$ and $N^{-(1+\alpha)}$, showing that threshold estimation benefits significantly from favorable margin conditions.

Overall, the result highlights that both regression estimation and threshold estimation become easier under the margin assumption, and that the availability of unlabeled data is particularly beneficial in this regime.

Remark 4 (Strongly proper losses)**.** *The bounds above are expressed in terms of the regression error $\|\eta - \hat{\eta}\|$, which is natural from a statistical perspective. In practice, however, predictors are obtained by minimizing surrogate losses. To bridge this gap, we rely on the theory of strongly proper composite losses. Let $\ell(y, f(x))$ be a λ-strongly proper composite loss with link function $\psi : [0, 1] \to \mathbb{R}$, and denote by $\psi^{-1} \circ f$ the induced probability estimate. The associated surrogate regret (i.e., excess risk) is defined as*

$$\mathrm{regret}_\ell[f] := \mathbb{E}[\ell(Y, f(X))] - \inf_{f'} \mathbb{E}[\ell(Y, f'(X))].$$

Such losses provide calibrated estimates of η, in the sense that

$$\mathbb{E}\left[(\eta(X) - \psi^{-1}(f(X)))^2\right] \leqslant \frac{2}{\lambda}\,\mathrm{regret}_\ell[f], \tag{13}$$

see, e.g., Agarwal (2014); Kotlowski & Dembczyński (2016). This shows that the regression error term in our bounds can be controlled directly by the surrogate regret. As a consequence, our framework applies to any algorithm minimizing a strongly proper loss, yielding a black-box reduction from excess utility under linear-fractional metrics to surrogate regret.

5 Experiments

5.1 Simulations

We evaluate our theoretical findings on two synthetic settings. We compare the classifier in Eq. (8) with three thresholding baselines: a fixed threshold $\hat{\theta} = 0.5$, a threshold selected by 10-fold cross-validation on labeled data, and an oracle threshold θ^* obtained by fine-grid search using the true regression function η.

Logistic DGP. We draw $X \sim \mathcal{N}(0, \mathrm{I}_d)$ with $d = 10$, set $\eta(x) = \sigma(aw^\top x + b)$ with $a = 2$, $b = -1.5$, and a fixed unit vector $w \in \mathbb{R}^d$, and generate labels as $Y \mid X \sim \mathrm{Bernoulli}(\eta(X))$.

This gives $\mathbb{P}(Y = 1) \approx 0.29$. Logistic regression is well specified in this setting, since the regression function belongs to the model class. We also consider a misspecified variant in which the data are generated from a probit link, $\eta(x) = \Phi(aw^\top x + b)$, while a logistic model is fitted. In both cases, we fix $n = 1000$ labeled observations and vary the number of unlabeled samples $N \in \{5, 10, 20, 40, 75, 100, 200, 500, 1000\}$ over 100 independent replications. We use the F_1-score as the utility.

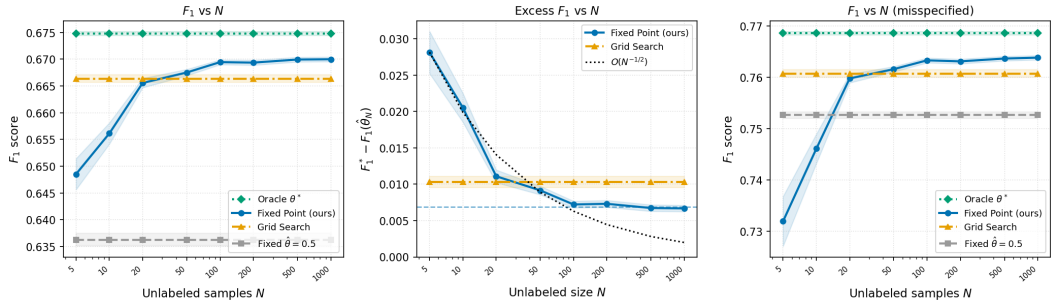


Figure 1: Logistic DGP. Left: F_1 score as a function of N in the well-specified setting. Center: excess risk as a function of N in the well-specified setting. Right: F_1 score as a function of N in the misspecified setting.

Gaussian mixture DGP. We draw $Y \sim \text{Bernoulli}(p)$ with $p = 0.3$ and $X \mid Y = k \sim \mathcal{N}(\mu_k, \mathrm{I}_d)$ with $d = 5$, $\mu_1 = (-0.5, \ldots, -0.5)^\top$, and $\mu_0 = (0.5, \ldots, 0.5)^\top$. Under equal isotropic covariances, the true regression function is available in closed form:

$$\eta(x) = \frac{p\,\phi_1(x)}{p\,\phi_1(x) + (1-p)\,\phi_0(x)}$$

where $\phi_k(x) = \mathcal{N}(x; \mu_k, I_d)$ denotes the Gaussian density of class k. In this setting, logistic regression is again well specified, since the log-odds are linear in x. The left panel of Figure 2 visualizes the two-class structure projected onto the first two coordinates. We fix $n = 1000$ labeled observations and vary $N \in \{10, 20, 50, 100, 200, 500, 1000, 2000\}$ over 100 independent replications. We use both the F_1-score and the Jaccard index as utilities.

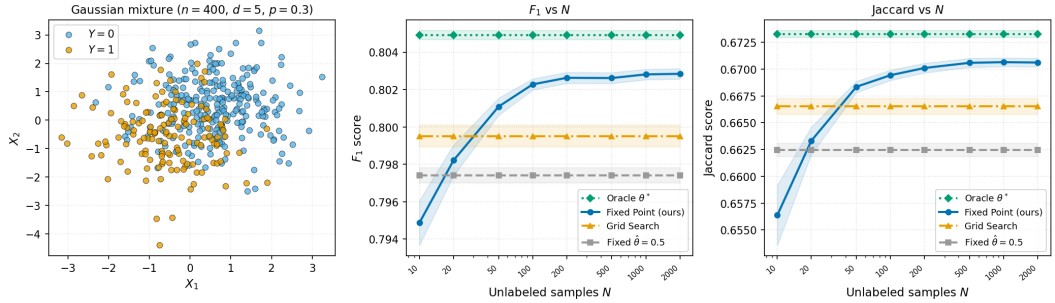


Figure 2: Gaussian mixture DGP. Left: class distribution projected onto the first two coordinates. Center: F_1 score as a function of N. Right: Jaccard index as a function of N.

Results. In both synthetic settings, our estimator improves as the number of unlabeled samples N increases, and outperforms the standard cross-validation baseline based on grid search. At the same time, the method does not converge exactly to the oracle as $N \to \infty$, because the remaining error is due to estimating the regression function, which depends on the number of labeled samples n rather than on N. The center panel of Figure 1 illustrates the predicted $O(N^{-1/2})$ behavior of the excess risk. The decrease is visible for small values of N, where the dominant contribution comes from threshold estimation, and then flattens once this term becomes negligible compared with the regression error $\mathbb{E}\|\eta - \hat{\eta}\|_1$. The Jaccard results in Figure 2 show the same qualitative behavior.

5.2 Real Data

We evaluate the methods on several real datasets; dataset descriptions and additional experimental details are provided in Appendix B. For each dataset, we use 80% of the data for training and 20% for testing. The training set is further split into a 50% portion used to estimate the regression function and a 50% portion used for threshold estimation. The fixed-threshold baseline with $\hat{\theta} = 0.5$ does not use the latter split, the cross-validation baseline uses both inputs and labels, while our bisection method uses only the inputs.

Dataset	$\hat{\eta}$	$\hat{\theta} = 0.5$		10-fold CV		Bisection (ours)	
		F_1	Time	F_1	Time	F_1	Time
German	LinReg	0.537 (0.059)	1.9	0.587 (0.056)	26.8	**0.592** (0.049)	2.4
	LR	0.401 (0.075)	6.4	0.586 (0.053)	30.0	0.579 (0.043)	6.8
	KRR	0.502 (0.090)	31.8	0.584 (0.045)	56.6	0.585 (0.045)	32.3
Spambase	LinReg	0.842 (0.014)	8.8	0.878 (0.012)	22.6	0.874 (0.012)	9.2
	LR	0.732 (0.020)	6.0	0.829 (0.017)	15.7	0.746 (0.019)	6.3
	KRR	0.898 (0.014)	460.3	0.902 (0.012)	473.7	**0.903** (0.011)	460.7
Sonar	LinReg	0.738 (0.108)	2.0	0.763 (0.094)	16.1	0.764 (0.105)	2.3
	LR	0.762 (0.104)	3.4	0.772 (0.092)	16.4	0.742 (0.086)	3.7
	KRR	0.869 (0.089)	3.3	0.864 (0.087)	18.0	**0.878** (0.079)	3.7
Phoneme	LinReg	0.520 (0.023)	1.2	0.649 (0.013)	15.4	0.640 (0.016)	1.6
	LR	0.399 (0.026)	8.2	0.635 (0.013)	21.2	0.631 (0.014)	8.6
	KRR	0.663 (0.021)	665.9	0.714 (0.017)	679.4	**0.718** (0.016)	666.2
Adult	LinReg	0.446 (0.014)	8.9	0.598 (0.007)	36.0	0.588 (0.007)	10.2
	LR	0.490 (0.011)	34.3	0.605 (0.006)	54.2	0.606 (0.007)	35.3
	KRR	0.592 (0.024)	232.6	**0.663** (0.008)	259.7	**0.663** (0.008)	234.0
Creditcard	LinReg	0.551 (0.048)	120.7	**0.813** (0.029)	190.5	0.768 (0.040)	125.3
	LR	0.298 (0.050)	166.6	0.662 (0.043)	235.5	0.434 (0.046)	171.1
	KRR	0.610 (0.232)	260.2	0.762 (0.185)	327.8	0.751 (0.167)	264.7
Mammogr.	LinReg	0.246 (0.073)	1.4	0.507 (0.068)	16.7	0.473 (0.057)	2.1
	LR	0.004 (0.011)	5.7	0.523 (0.108)	19.5	0.331 (0.111)	6.2
	KRR	0.440 (0.105)	108.5	0.608 (0.064)	123.6	**0.619** (0.054)	109.1

Table 2: F_1 mean (std) and runtime (ms) across datasets.

Results in Table 2 show that methods which calibrate the threshold consistently outperform the fixed-threshold ERM baseline, confirming the relevance of threshold selection when the target metric differs from accuracy. The bisection method performs comparably to cross-validation while being consistently faster. Its performance depends on the quality of the regression estimator, as expected from the fixed-point equation (6); the best results are typically obtained with KRR using a Gaussian kernel. Importantly, unlike cross-validation, the proposed bisection method does not require labels for threshold estimation, which can be a significant advantage in applications where labels are expensive.

As expected, the main computational bottleneck remains the estimation of the regression function. Since this is not the focus of our experiments, we use standard `scikit-learn` implementations. Faster regression methods, including approaches based on random projections and Nyström approximations, could be used instead (Meanti et al., 2020; Della Vecchia et al., 2021, 2024).

6 Conclusions and Future Directions

In this paper, we studied binary classification under linear-fractional performance metrics, a broad class of non-decomposable objectives commonly used in imbalanced settings. We showed that the optimal classifier admits a simple thresholding structure on the regression function, reducing the original infinite-dimensional problem to the estimation of a single scalar parameter characterized by a fixed-point equation. Building on this characterization, we analyzed a plug-in approach that

decouples regression estimation from threshold estimation and leverages unlabeled data to estimate the optimal threshold. Our finite-sample guarantees provide a modular decomposition of the excess utility, isolating the contributions of regression error, threshold estimation, and sampling effects, and showing how each component can be controlled independently.

Many directions for future work remain open. In particular, our framework naturally connects to domain adaptation and covariate shift (Sugiyama et al., 2007; Della Vecchia et al., 2026). Since the optimal threshold depends only on the marginal distribution of the inputs, it can be re-estimated using unlabeled data from the target domain, providing a simple mechanism to adapt to distribution shifts without retraining the regression function. More broadly, our results suggest extensions to richer prediction settings where non-decomposable metrics play a central role, such as multi-class classification and constrained learning (Narasimhan et al., 2024). Another promising direction concerns robustness to imperfect supervision, including settings with noisy or corrupted labels (Natarajan et al., 2018; Menon et al., 2015), where separating regression estimation from post-processing may provide additional stability. It would also be interesting to study extensions to decision problems with rejection options (Charoenphakdee et al., 2021) or class-weighted objectives (Xu et al., 2020), where thresholding rules naturally arise and trade-offs between different types of errors are explicitly controlled. Plug-in thresholding rules also arise in sequential decision problems such as optimal stopping, where the exercise decision compares a payoff with a regression estimate of the continuation value; extending our post-processing analysis to this setting, where regression errors propagate backward through the dynamic programming recursion Della Vecchia & Filipović (2025), is another natural direction.

References

Agarwal, S. Surrogate regret bounds for bipartite ranking via strongly proper losses. The Journal of Machine Learning Research, 15(1):1653–1674, 2014.

Audibert, J.-Y. and Tsybakov, A. B. Fast learning rates for plug-in classifiers. The Annals of statistics, 35(2):608–633, 2007.

Bao, H., Scott, C., and Sugiyama, M. Calibrated surrogate losses for adversarially robust classification. In Conference on Learning Theory, pp. 408–451. PMLR, 2020.

Berman, M., Triki, A. R., and Blaschko, M. B. The lovász-softmax loss: A tractable surrogate for the optimization of the intersection-over-union measure in neural networks. In Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 4413–4421, 2018.

Brodersen, K. H., Ong, C. S., Stephan, K. E., and Buhmann, J. M. The balanced accuracy and its posterior distribution. In 2010 20th international conference on pattern recognition, pp. 3121–3124. IEEE, 2010.

Charoenphakdee, N., Lee, J., and Sugiyama, M. On symmetric losses for learning from corrupted labels. In International Conference on Machine Learning, pp. 961–970. PMLR, 2019.

Charoenphakdee, N., Cui, Z., Zhang, Y., and Sugiyama, M. Classification with rejection based on cost-sensitive classification. In International Conference on Machine Learning, pp. 1507–1517. PMLR, 2021.

Chzhen, E. Optimal rates for nonparametric f-score binary classification via post-processing. Mathematical Methods of Statistics, 29(2):87–105, 2020.

Della Vecchia, A. and Filipović, D. Error propagation in dynamic programming: From stochastic control to option pricing. arXiv preprint arXiv:2509.20239, 2025.

Della Vecchia, A., Mourtada, J., De Vito, E., and Rosasco, L. Regularized erm on random subspaces. In International Conference on Artificial Intelligence and Statistics, pp. 4006–4014. PMLR, 2021.

Della Vecchia, A., De Vito, E., Mourtada, J., and Rosasco, L. The nyström method for convex loss functions. Journal of Machine Learning Research, 25(360):1–60, 2024.

Della Vecchia, A., Watusadisi, A. M., De Vito, E., and Rosasco, L. Computational efficiency under covariate shift in kernel ridge regression. Advances in Neural Information Processing Systems, 38:141618–141648, 2026.

Dembczyński, K., Kotłowski, W., Koyejo, O., and Natarajan, N. Consistency analysis for binary classification revisited. In International Conference on Machine Learning, pp. 961–969. PMLR, 2017.

Drummond, C. and Holte, R. C. Severe class imbalance: Why better algorithms aren't the answer. In European Conference on Machine Learning, pp. 539–546. Springer, 2005.

Gaucher, S., Schreuder, N., and Chzhen, E. Fair learning with wasserstein barycenters for non-decomposable performance measures. In International Conference on Artificial Intelligence and Statistics, pp. 2436–2459. PMLR, 2023.

Geer, S. A. Empirical Processes in M-estimation, volume 6. Cambridge university press, 2000.

Gu, Q., Zhu, L., and Cai, Z. Evaluation measures of the classification performance of imbalanced data sets. In International symposium on intelligence computation and applications, pp. 461–471. Springer, 2009.

He, H. and Garcia, E. A. Learning from imbalanced data. IEEE Transactions on knowledge and data engineering, 21(9):1263–1284, 2009.

Joachims, T. A support vector method for multivariate performance measures. In Proceedings of the 22nd international conference on Machine learning, pp. 377–384, 2005.

Kotlowski, W. and Dembczyński, K. Surrogate regret bounds for generalized classification performance metrics. In Asian Conference on Machine Learning, pp. 301–316. PMLR, 2016.

Koyejo, O., Natarajan, N., Ravikumar, P., and Dhillon, I. S. Consistent binary classification with generalized performance metrics. Advances in neural information processing systems, 27, 2014.

Lewis, D. D. A sequential algorithm for training text classifiers: Corrigendum and additional data. In Acm sigir forum, volume 29, pp. 13–19. ACM New York, NY, USA, 1995.

Meanti, G., Carratino, L., Rosasco, L., and Rudi, A. Kernel methods through the roof: handling billions of points efficiently. Advances in Neural Information Processing Systems, 33:14410–14422, 2020.

Menon, A., Van Rooyen, B., Ong, C. S., and Williamson, B. Learning from corrupted binary labels via class-probability estimation. In International conference on machine learning, pp. 125–134. PMLR, 2015.

Narasimhan, H., Vaish, R., and Agarwal, S. On the statistical consistency of plug-in classifiers for non-decomposable performance measures. Advances in neural information processing systems, 27, 2014.

Narasimhan, H., Kar, P., and Jain, P. Optimizing non-decomposable performance measures: A tale of two classes. In International Conference on Machine Learning, pp. 199–208. PMLR, 2015.

Narasimhan, H., Ramaswamy, H. G., Tavker, S. K., Khurana, D., Netrapalli, P., and Agarwal, S. Consistent multiclass algorithms for complex metrics and constraints. Journal of Machine Learning Research, 25(367):1–81, 2024.

Natarajan, N., Dhillon, I. S., Ravikumar, P., and Tewari, A. Cost-sensitive learning with noisy labels. Journal of Machine Learning Research, 18(155):1–33, 2018.

Parambath, S. A., Usunier, N., and Grandvalet, Y. Optimizing f-measures by cost-sensitive classification. Advances in neural information processing systems, 27, 2014.

Sugiyama, M., Krauledat, M., and Müller, K.-R. Covariate shift adaptation by importance weighted cross validation. Journal of Machine Learning Research, 8(5), 2007.

Xu, Z., Dan, C., Khim, J., and Ravikumar, P. Class-weighted classification: Trade-offs and robust approaches. In International conference on machine learning, pp. 10544–10554. PMLR, 2020.

Yan, B., Koyejo, S., Zhong, K., and Ravikumar, P. Binary classification with karmic, threshold-quasi-concave metrics. In International Conference on Machine Learning, pp. 5531–5540. PMLR, 2018.

Ye, N., Chai, K. M., Lee, W. S., and Chieu, H. L. Optimizing f-measures: A tale of two approaches. In Proceedings of the 29th International Conference on Machine Learning, pp. 289–296. Omnipress, 2012.

A Main Proofs

We report here the proofs of the main results in the paper.

Proof of Lemma 1. The idea is to apply the first-order optimality condition to $\mathrm{U}(g_\theta) = \mathrm{U}(\theta)$. We start noting that

$$\begin{aligned}\mathbb{P}[Y=1, g(X)=1] = \mathbb{E}[Yg(X)] &= \int_{\mathcal{X}} \eta(x)\mathbb{1}\{\eta(x) \geqslant \theta\}\mathrm{d}\mathbb{P}_X(x) \\ &= \int_\theta^1 z\mathrm{d}F_Z(z) = \int_{F_Z(\theta)}^{F_Z(1)} F_Z^{-1}(u)\mathrm{d}u, \end{aligned} \tag{14}$$

where we took $Z = \eta(X)$ and we used the fact that $\mathbb{E}_{X\sim F_X}[X] = \mathbb{E}_{U\sim\text{Unif}[0,1]}[F^{-1}(U)]$. Differentiating (14) wrt θ we obtain

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\int_{F_{\eta(X)}(\theta)}^{F_{\eta(X)}(1)} F_{\eta(X)}^{-1}(u)\mathrm{d}u\right) = -F_{\eta(X)}^{-1}(F_{\eta(X)}(\theta))F'_{\eta(X)}(\theta) = -\theta f_{\eta(X)}(\theta),$$

with $f_{\eta(X)}(\cdot)$ the density function. Similarly, we have

$$\mathbb{P}[g(X)=1] = \mathbb{P}[\eta(X) \geqslant \theta] = 1 - F_{\eta(X)}(\theta)$$

and

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(1 - F_{\eta(X)}(\theta)\right) = -f_{\eta(X)}(\theta).$$

Then, applying the first order optimality condition to $\mathrm{U}(\theta)$ we obtain the following equation

$$\begin{aligned}&-f_{\eta(X)}(\theta)(c_1\theta + c_2)(d_0 + d_1\mathbb{E}[(\eta(X)-\theta)_+] + d_1\theta(1-F_{\eta(X)}(\theta)) + d_2(1-F_{\eta(X)}(\theta))) + \\ &+f_{\eta(X)}(\theta)(d_1\theta + d_2)(c_0 + c_1\mathbb{E}[(\eta(X)-\theta)_+] + c_1\theta(1-F_{\eta(X)}(\theta)) + c_2(1-F_{\eta(X)}(\theta))) = 0.\end{aligned}$$

Rearranging and simplifying the various terms we get the desired condition on θ^*:

$$\mathbb{E}[(\eta(X)-\theta^*)_+] = \frac{c_0d_1 - c_1d_0}{c_2d_1 - c_1d_2}\theta^* + \frac{c_0d_2 - c_2d_0}{c_2d_1 - c_1d_2} \tag{15}$$

where we used $\mathbb{E}[Yg(X)] = \mathbb{E}[\eta(X)\mathbb{1}\{\eta(X) \geqslant \theta\}] = \mathbb{E}[(\eta(X)-\theta)_+] + \theta\mathbb{P}[\eta(X) \geqslant \theta]$. □

Proof of Lemma 3. Using the optimality condition (4) for θ^* we have

$$\begin{aligned}\mathbb{E}[\eta(X)\mathbb{1}\{\eta(X) \geqslant \theta^*\}] &= \mathbb{E}[(\eta(X)-\theta^*)_+] + \theta^*\mathbb{P}[\eta(X) \geqslant \theta^*] \\ &= \frac{c_0d_1 - c_1d_0}{c_2d_1 - c_1d_2}\theta^* + \frac{c_0d_2 - c_2d_0}{c_2d_1 - c_1d_2} + \theta^*\mathbb{P}[\eta(X) \geqslant \theta^*]. \end{aligned} \tag{16}$$

Then, we can rewrite $\mathrm{U}_{(c,d)}(g_{\theta^*})$ as

$$\mathrm{U}_{(c,d)}(g_{\theta^*}) = \frac{c_0(c_2d_1 - d_2c_1) + c_1(\theta^*(c_0d_1 - d_0c_1) + (c_0d_2 - d_0c_2)) + (c_2 + \theta^*c_1)(c_2d_1 - d_2c_1)\mathbb{P}[\eta(X) \geqslant \theta^*]}{d_0(c_2d_1 - d_2c_1) + d_1(\theta^*(c_0d_1 - d_0c_1) + (c_0d_2 - d_0c_2)) + (d_2 + \theta^*d_1)(c_2d_1 - d_2c_1)\mathbb{P}[\eta(X) \geqslant \theta^*]}$$

Simplifying some terms and factorizing the numerator and the denominator by $(c_2 + \theta^*c_1)$ and $(d_2 + \theta^*d_1)$ respectively, we obtain

$$\mathrm{U}_{(c,d)}(g_{\theta^*}) = \frac{c_2 + \theta^*c_1}{d_2 + \theta^*d_1} \cdot \frac{(c_0d_1 - d_0c_1) + (c_2d_1 - d_2c_1)\mathbb{P}[\eta(X) \geqslant \theta^*]}{(c_0d_1 - d_0c_1) + (c_2d_1 - d_2c_1)\mathbb{P}[\eta(X) \geqslant \theta^*]} = \frac{c_2 + \theta^*c_1}{d_2 + \theta^*d_1}.$$

□

Proof of Lemma 2. Let's call $D(g_{\theta^*}) := d_0 + d_1\mathbb{E}[\eta(X)\mathbb{1}\{\eta(x) \geqslant \theta^*\}] + d_2\mathbb{P}[\eta(x) \geqslant \theta^*]$ and $D(g) := d_0 + d_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + d_2\mathbb{P}[g(X) = 1]$, we have

$$
\begin{aligned}
&\mathrm{U}(g_{\theta^*}) - \mathrm{U}(g) = \\
&= \frac{c_0 + c_1\mathbb{E}[\eta(X)\mathbb{1}\{\eta(x) \geqslant \theta^*\}] + c_2\mathbb{P}[\eta(x) \geqslant \theta^*]}{D(g_{\theta^*})} - \frac{c_0 + c_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + c_2\mathbb{P}[g(X) = 1]}{D(g)} \\
&= \frac{c_1\mathbb{E}[(\eta(X) - \theta^*)(\mathbb{1}\{\eta(x) \geqslant \theta^*\} - \mathbb{1}\{g(X) = 1\})] + (c_1\theta^* + c_2)(\mathbb{P}[\eta(x) \geqslant \theta^*] - \mathbb{P}[g(X) = 1])}{D(g_{\theta^*})} \\
&+ \frac{(c_0 + c_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + c_2\mathbb{P}[g(X) = 1])(d_0 + d_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + d_2\mathbb{P}[g(X) = 1])}{D(g)D(g_{\theta^*})} \\
&- \frac{(c_0 + c_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + c_2\mathbb{P}[g(X) = 1])(d_0 + d_1\mathbb{E}[\eta(X)\mathbb{1}\{\eta(x) \geqslant \theta^*\}] + d_2\mathbb{P}[\eta(x) \geqslant \theta^*])}{D(g)D(g_{\theta^*})} \\
&= \frac{c_1\mathbb{E}[(\eta(X) - \theta^*)(\mathbb{1}\{\eta(x) \geqslant \theta^*\} - \mathbb{1}\{g(X) = 1\})] + (c_1\theta^* + c_2)(\mathbb{P}[\eta(X) \geqslant \theta^*] - \mathbb{P}[g(X) = 1])}{D(g_{\theta^*})} + \\
&\quad + \frac{(c_0 + c_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + c_2\mathbb{P}[g(X) = 1])(d_1\mathbb{E}[\eta(X)(\mathbb{1}\{g(X) = 1\} - \mathbb{1}\{\eta(X) \geqslant \theta^*\})]}{D(g)D(g_{\theta^*})} \\
&\quad + \frac{d_2(\mathbb{P}[g(X) = 1] - \mathbb{P}[\eta(X) \geqslant \theta^*]))}{D(g)D(g_{\theta^*})} \\
&= \frac{c_1\mathbb{E}[(\eta(X) - \theta^*)(\mathbb{1}\{\eta(X) \geqslant \theta^*\} - \mathbb{1}\{g(X) = 1\})] + (c_1\theta^* + c_2)(\mathbb{P}[\eta(X) \geqslant \theta^*] - \mathbb{P}[g(X) = 1])}{D(g_{\theta^*})} \\
&\quad + \mathrm{U}(g)\frac{d_1\mathbb{E}[\eta(X)(\mathbb{1}\{g(X) = 1\} - \mathbb{1}\{\eta(X) \geqslant \theta^*\})] + d_2(\mathbb{P}[g(X) = 1] - \mathbb{P}[\eta(X) \geqslant \theta^*])}{D(g_{\theta^*})} \\
&= \frac{c_1\mathbb{E}[|\eta(X) - \theta^*|\mathbb{1}\{g_{\theta^*}(X) \neq g(X)\}]}{D(g_{\theta^*})} + \mathrm{U}(g)\frac{d_1\mathbb{E}[\eta(X)(\mathbb{1}\{g(X) = 1\} - \mathbb{1}\{\eta(X) \geqslant \theta^*\})]}{D(g_{\theta^*})} \\
&\quad + (c_1\theta^* + c_2 - d_2\mathrm{U}(g))\frac{\mathbb{P}[\eta(X) \geqslant \theta^*] - \mathbb{P}[g(X) = 1]}{D(g_{\theta^*})} \\
&= \frac{c_1\mathbb{E}[|\eta(X) - \theta^*|\mathbb{1}\{g_{\theta^*}(X) \neq g(X)\}]}{D(g_{\theta^*})} \\
&\quad + d_1(\mathrm{U}(g_{\theta^*}) - \mathrm{U}(g))\frac{\mathbb{E}[\eta(X)(\mathbb{1}\{\eta(X) \geqslant \theta^*\} - \mathbb{1}\{g(X) = 1\})]}{D(g_{\theta^*})} \\
&\quad + d_2(\mathrm{U}(g_{\theta^*}) - \mathrm{U}(g))\frac{\mathbb{P}[\eta(x) \geqslant \theta^*] - \mathbb{P}[g(X) = 1]}{D(g_{\theta^*})},
\end{aligned}
$$

where we added and subtracted $\frac{c_0 + c_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X)=1\}] + c_2\mathbb{P}[g(X)=1]}{D(g_{\theta^*})}$ and we used the fact that $\mathrm{U}(g_{\theta^*}) = \frac{c_2 + \theta^* c_1}{d_2 + \theta^* d_1}$ (see Lemma 3). Solving for U we get

$$
\mathrm{U}(g_{\theta^*}) - \mathrm{U}(g) = \frac{c_1\mathbb{E}[|\eta(X) - \theta^*|\mathbb{1}\{g_{\theta^*}(X) \neq g(X)\}]}{d_0 + d_1\mathbb{E}[\eta(X)\mathbb{1}\{g(X) = 1\}] + d_2\mathbb{P}[g(X) = 1]},
$$

that is non-negative under Assumption 2. □

Proof of Proposition 1. We already know we can write the excess utility as

$$
\mathrm{U}(g_{\theta^*}) - \mathrm{U}(\hat{g}) = \frac{c_1\mathbb{E}[|\eta(X) - \theta^*|\mathbb{1}\{g_{\theta^*}(X) \neq \hat{g}(X)\}]}{d_0 + d_1\mathbb{E}[\eta(X)\mathbb{1}\{\hat{g}(X) = 1\}] + d_2\mathbb{P}[\hat{g}(X) = 1]}.
$$

We can rewrite the numerator as:

$$
\begin{aligned}
&\mathbb{E}[|\eta(X) - \theta^*|\mathbb{1}\{g_{\theta^*}(X) \neq \hat{g}(X)\}] = \\
&\quad = \mathbb{E}[(\eta(X) - \theta^*)\mathbb{1}\{\eta(X) \geqslant \theta^*, \hat{\eta}(X) < \hat{\theta}\}] + \mathbb{E}[(\theta^* - \eta(X))\mathbb{1}\{\eta(X) < \theta^*, \hat{\eta}(X) \geqslant \hat{\theta}\}] \\
&\quad \leqslant \mathbb{E}[(\eta(X) - \theta^* - \hat{\eta}(X) + \hat{\theta})\mathbb{1}\{\eta(X) \geqslant \theta^*, \hat{\eta}(X) < \hat{\theta}\}] + \\
&\qquad + \mathbb{E}[(\theta^* - \eta(X) - \hat{\theta} + \hat{\eta}(X))\mathbb{1}\{\eta(X) < \theta^*, \hat{\eta}(X) \geqslant \hat{\theta}\}] \\
&\quad \leqslant \|\eta - \hat{\eta}\|_1 + |\theta^* - \hat{\theta}|.
\end{aligned}
$$

Then we obtain

$$\mathrm{U}(g_{\theta^*}) - \mathrm{U}(\hat{g}) \leqslant c_1 \frac{\|\eta - \hat{\eta}\|_1 + |\theta^* - \hat{\theta}|}{d_0 + d_1 \mathbb{E}[\eta(X)\mathbb{1}\{\hat{g}(X) = 1\}] + d_2 \mathbb{P}[\hat{g}(X) = 1]}.$$

We start by controlling $|\theta^* - \hat{\theta}|$. Define $\bar{\theta}$ such that

$$\hat{R}(\bar{\theta}) = 0, \qquad \text{with} \quad \hat{R}(\theta) = \frac{c_0 d_1 - c_1 d_0}{c_2 d_1 - c_1 d_2}\theta + \frac{c_0 d_2 - c_2 d_0}{c_2 d_1 - c_1 d_2} - \hat{\mathbb{E}}_N(\hat{\eta}(X) - \theta)_+, \tag{17}$$

with $\hat{\theta}$ an estimator of $\bar{\theta}$ (for instance the output of a bisection algorithm applied to $\hat{R}(\theta)$) and $\hat{\mathbb{E}}_N$ is an expectation taken with respect to the empirical measure $\frac{1}{N}\sum_{i=n+1}^{n+N} \delta_{X_i}$ evaluated on unlabelled data.
We can decompose

$$\mathbb{E}_{\mathcal{D}_N^U}|\theta^* - \hat{\theta}| \leqslant \mathbb{E}_{\mathcal{D}_N^U}|\bar{\theta} - \hat{\theta}| + \mathbb{E}_{\mathcal{D}_N^U}|\bar{\theta} - \theta^*|.$$

The first term is simply an optimization error related with the chosen algorithm to solve $\hat{R}(\theta) = 0$. Notice that $\hat{R}(\theta)$ is continuous on $[0, 1]$ and $\hat{R}(0) \leqslant 0$, $\hat{R}(1) \geqslant 0$ thanks to the assumption in Eq. (7), then classical analysis of the bisection algorithm implies that $|\bar{\theta} - \hat{\theta}| \leqslant 2^{-K_{max}}$ almost surely.
To bound $|\bar{\theta} - \theta^*|$ we use Chzhen (2020).

$$|\bar{\theta} - \theta^*| = \frac{c_2 d_1 - c_1 d_2}{c_0 d_1 - c_1 d_0} \left| \int_{\theta^*}^1 (1 - F_\eta(t))dt - \int_{\bar{\theta}}^1 (1 - \hat{F}_{\hat{\eta}}(t))dt \right|. \tag{18}$$

Consider $\theta^* \geqslant \bar{\theta}$:

$$\left| \int_{\theta^*}^1 (1 - F_\eta(t))dt - \int_{\bar{\theta}}^1 (1 - \hat{F}_{\hat{\eta}}(t))dt \right| \leqslant \left| \int_{\theta^*}^1 (1 - F_\eta(t))dt - \int_{\theta^*}^1 (1 - \hat{F}_{\hat{\eta}}(t))dt \right|$$
$$\leqslant \left| \int_{\theta^*}^1 (\hat{F}_{\hat{\eta}}(t) - F_\eta(t))dt \right| \leqslant \int_{\theta^*}^1 \left| \hat{F}_{\hat{\eta}}(t) - F_\eta(t) \right| dt \leqslant \int_0^1 \left| \hat{F}_{\hat{\eta}}(t) - F_\eta(t) \right| dt \leqslant \|\hat{F}_{\hat{\eta}} - F_\eta\|_1.$$

Consider $\theta^* \leqslant \bar{\theta}$:

$$\left| \int_{\theta^*}^1 (1 - F_\eta(t))dt - \int_{\bar{\theta}}^1 (1 - \hat{F}_{\hat{\eta}}(t))dt \right| \leqslant \left| \int_{\bar{\theta}}^1 (1 - F_\eta(t))dt - \int_{\bar{\theta}}^1 (1 - \hat{F}_{\hat{\eta}}(t))dt \right|$$
$$\leqslant \left| \int_{\bar{\theta}}^1 (\hat{F}_{\hat{\eta}}(t) - F_\eta(t))dt \right| \leqslant \int_{\bar{\theta}}^1 \left| \hat{F}_{\hat{\eta}}(t) - F_\eta(t) \right| dt \leqslant \int_0^1 \left| \hat{F}_{\hat{\eta}}(t) - F_\eta(t) \right| dt \leqslant \|\hat{F}_{\hat{\eta}} - F_\eta\|_1.$$

So we have

$$|\bar{\theta} - \theta^*| = \frac{c_2 d_1 - c_1 d_2}{c_0 d_1 - c_1 d_0} \left| \int_{\theta^*}^1 (1 - F_\eta(t))dt - \int_{\bar{\theta}}^1 (1 - \hat{F}_{\hat{\eta}}(t))dt \right| \leqslant \frac{c_2 d_1 - c_1 d_2}{c_0 d_1 - c_1 d_0} \|\hat{F}_{\hat{\eta}} - F_\eta\|_1.$$

Now we can introduce also $\hat{F}_\eta$ that is the empirical cumulative distribution of $\eta(X)$ based on $\mathcal{D}_N^U$. Using the triangular inequality again we get

$$\|\hat{F}_{\hat{\eta}} - F_\eta\|_1 \leqslant \|F_\eta - \hat{F}_\eta\|_1 + \|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\|_1.$$

Following the reasoning in Chzhen (2020), we can concentrate the first term using Bernstein's inequality. Let $p(t) = P(\eta(X) \geqslant t)$, we have:

$$\mathbb{E}_{\mathcal{D}_N^U} \left| P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t) \right| = \int_0^\infty P_{\mathcal{D}_N^U} \left(\left| P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t) \right| \geqslant x \right) dx$$
$$\leqslant 2 \int_0^\infty \exp\left(-\frac{Nx^2}{2\left(p(t) + \frac{1}{3}x\right)} \right) dx,$$

with $\hat{P}_N$ the empirical measure $\frac{1}{N}\sum_{i=n+1}^{n+N} \delta_{X_i}$ evaluated on the unlabelled data.
Furthermore, we can write for the inner integral

$$\int_0^\infty \exp\left(-\frac{Nx^2}{2\left(p(t) + \frac{1}{3}x\right)} \right) dx = \left(\int_0^{3p(t)} + \int_{3p(t)}^\infty \right) \exp\left(-\frac{Nx^2}{2\left(p(t) + \frac{1}{3}x\right)} \right) dx$$
$$\leqslant \int_0^{3p(t)} \exp\left(-\frac{Nx^2}{4p(t)} \right) dx + \int_{3p(t)}^\infty \exp\left(-\frac{Nx}{4} \right) dx$$
$$\leqslant \sqrt{\frac{\pi p(t)}{N}} + \frac{4}{N}.$$

Therefore, using Fubini's theorem, we obtain

$$\mathbb{E}_{\mathcal{D}_N^U}\left\|F_\eta - \hat{F}_\eta\right\|_1 = \int_0^1 \mathbb{E}_{\mathcal{D}_N^U}\left|P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t)\right| dt$$
$$\leqslant \sqrt{\frac{\pi}{N}} \int_0^1 \sqrt{P(\eta(X) \geqslant t)} dt + \frac{4}{N}.$$

Applying Cauchy-Schwarz inequality to the first term on the r.h.s. of the above inequality we derive the following bound

$$\mathbb{E}_{\mathcal{D}_N^U}\left\|F_\eta - \hat{F}_\eta\right\|_1 \leqslant \sqrt{\frac{\pi\mathbb{P}(Y=1)}{N}} + \frac{4}{N}.$$

It remains to bound $\left\|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\right\|_1$. Let $Z_i = \eta(X_i)$ and $\hat{Z}_i = \hat{\eta}(X_i)$ for all $i = n+1, \ldots, n+N$, then we can write

$$\left\|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\right\|_1 = \frac{1}{N}\int_0^1 \left|\sum_{i=n+1}^{n+N}\left(\mathbb{1}\{Z_i \leqslant t\} - \mathbb{1}\left\{\hat{Z}_i \leqslant t\right\}\right)\right| dt.$$

This expression corresponds to the Wasserstein-1 distance between empirical measures of $\{Z_{n+1}, \ldots, Z_{n+N}\}$ and $\left\{\hat{Z}_{n+1}, \ldots, \hat{Z}_{n+N}\right\}$. Hence, using its alternative definition we get

$$\left\|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\right\|_1 = \inf_{\omega\in\mathbb{S}_N}\frac{1}{N}\sum_{i=n+1}^{n+N}\left|Z_i - \hat{Z}_{\omega(i)}\right| \leqslant \frac{1}{N}\sum_{i=n+1}^{n+N}\left|\eta(X_i) - \hat{\eta}(X_i)\right|,$$

where the infimum is taken over all permutations $\mathbb{S}_N$ of $\{n+1, \ldots, n+N\}$. Finally, since conditionally on the labelled data $\mathcal{D}_n^{\mathrm{L}}$ the random variables $|\eta(X_i) - \hat{\eta}(X_i)|$ with $i = n+1, \ldots, n+N$ are i.i.d., then $\mathbb{E}_{\mathcal{D}_N^U}\left\|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\right\|_1 \leqslant \mathbb{E}_{\mathcal{D}_N^U}\|\eta - \hat{\eta}\|_1$. Hence,

$$\mathbb{E}_{\mathcal{D}_N^U}\|\hat{F}_{\hat{\eta}} - F_\eta\|_1 \leqslant \sqrt{\frac{\pi\mathbb{P}(Y=1)}{N}} + \frac{4}{N} + \|\eta - \hat{\eta}\|_1.$$

Putting all together

$$\mathbb{E}_{\mathcal{D}_N^U}|\theta^* - \hat{\theta}| \leqslant \frac{c_2 d_1 - c_1 d_2}{c_0 d_1 - c_1 d_0}\left(\sqrt{\frac{\pi\mathbb{P}(Y=1)}{N}} + \frac{4}{N} + \|\eta - \hat{\eta}\|_1\right) + 2^{-K_{\max}}. \tag{19}$$

As regards the denominator, conditions in Assumption 2 assure that $d_0 \geqslant 0$ and $d_2 \geqslant 0$. Then we get

$$d_0 + d_1\mathbb{P}[Y=1, \hat{g}(X)=1] + d_2\mathbb{P}[\hat{g}(X)=1] \geqslant d_0 + (d_1+d_2)\mathbb{P}[Y=1, \hat{g}(X)=1]$$
$$\geqslant d_0 + \min\{d_1+d_2, 0\}\mathbb{P}[Y=1].$$

So, with $K_0(\mathbb{P}, d_0, d_1) = 1/(d_0 + \min\{d_1+d_2, 0\}\mathbb{P}[Y=1])$, we obtain the final bound

$$\mathrm{U}(g_{\theta^*}) - \mathbb{E}_{\mathcal{D}_N^U}\mathrm{U}(\hat{g}) \leqslant c_1\frac{\|\eta - \hat{\eta}\|_1 + \mathbb{E}_{\mathcal{D}_N^U}|\theta^* - \hat{\theta}|}{d_0 + d_1\mathbb{P}[Y=1, \hat{g}(X)=1] + d_2\mathbb{P}[\hat{g}(X)=1]}$$
$$\leqslant c_1 K_0(\mathbb{P}, d_0, d_1)\left(\|\eta - \hat{\eta}\|_1 + \frac{c_2 d_1 - c_1 d_2}{c_0 d_1 - c_1 d_0}\left(\sqrt{\frac{\pi p}{N}} + \frac{4}{N} + \|\eta - \hat{\eta}\|_1\right) + 2^{-K_{\max}}\right).$$

Finally, taking the expectation $\mathbb{E}_{\mathcal{D}_n^L}$ over the labelled dataset we obtain the results. □

We introduce the following lemma we will need in the following.

Lemma 4. *On the set* $\{X : \hat{g}(X) \neq g_{\theta^*}\}$ *we have*

$$|\eta(X) - \theta^*| \leqslant 2\max\{|\eta(X) - \hat{\eta}(X)|, |\hat{\theta} - \theta^*|\}. \tag{20}$$

Proof of Lemma 4. We analyse the 2 possible cases:

- $\hat{\theta} < \theta^*$, $\eta(X) \geqslant \theta^*$, $\hat{\eta}(X) < \hat{\theta}$ (analogous to the symmetric case $\hat{\theta} \geqslant \theta^*$, $\eta(X) < \theta^*$, $\hat{\eta}(X) \geqslant \hat{\theta}$): then it's easy to see that $|\eta(X) - \theta^*| \leqslant |\eta(X) - \hat{\eta}(X)|$;
- $\hat{\theta} < \theta^*$, $\eta(X) < \theta^*$, $\hat{\eta}(X) \geqslant \hat{\theta}$ (analogous to the symmetric case $\hat{\theta} \geqslant \theta^*$, $\eta(X) \geqslant \theta^*$, $\hat{\eta}(X) < \hat{\theta}$): we have $|\eta(X) - \theta^*| \leqslant |\eta(X) - \hat{\eta}(X)| + |\hat{\theta} - \theta^*|$.

Putting all together we obtain the result. □

We can now prove Proposition 2.

Proof of Proposition 2. By Lemma 2 and the lower bound on the denominator used in Proposition 1, we have

$$\mathrm{U}(g_{\theta^*}) - \mathrm{U}(\hat{g}) \leqslant c_1 K_0 \, \mathbb{E}\big[|\eta(X) - \theta^*| \mathbb{1}\{g_{\theta^*}(X) \neq \hat{g}(X)\}\big].$$

Thus, it remains to control the margin-weighted disagreement term. By Lemma 4, on the disagreement set

$$\{g_{\theta^*}(X) \neq \hat{g}(X)\},$$

we have

$$|\eta(X) - \theta^*| \leqslant 2 \max\{|\eta(X) - \hat{\eta}(X)|, |\hat{\theta} - \theta^*|\}.$$

Hence,

$$\begin{aligned}
\frac{\mathrm{U}(g_{\theta^*}) - \mathrm{U}(\hat{g})}{c_1 K_0(\mathbb{P}, d_0, d_1)} &\leqslant \mathbb{E}[|\eta(X) - \theta^*| \mathbb{1}\{g_{\theta^*}(X) \neq \hat{g}(X)\}] \\
&\leqslant \mathbb{E}[|\eta(X) - \theta^*| \mathbb{1}\{|\eta(X) - \theta^*| \leqslant 2|\eta(X) - \hat{\eta}(X)|\}] + \\
&\quad + \mathbb{E}[|\eta(X) - \theta^*| \mathbb{1}\{|\eta(X) - \theta^*| \leqslant 2|\hat{\theta} - \theta^*|\}] \\
&\leqslant \mathbb{E}[|\eta(X) - \theta^*| \mathbb{1}\{|\eta(X) - \theta^*| \leqslant 2|\eta(X) - \hat{\eta}(X)|\} \mathbb{1}\{|\eta(X) - \theta^*| \leqslant t\}] + \\
&\quad + \mathbb{E}[|\eta(X) - \theta^*| \mathbb{1}\{|\eta(X) - \theta^*| \leqslant 2|\eta(X) - \hat{\eta}(X)|\} \mathbb{1}\{|\eta(X) - \theta^*| > t\}] + \\
&\quad + 2\mathbb{E}[|\hat{\theta} - \theta^*| \mathbb{1}\{|\eta(X) - \theta^*| \leqslant 2|\hat{\theta} - \theta^*|\}] \\
&\leqslant 2\, \mathbb{E}\big[|\eta(X) - \hat{\eta}(X)| \, \mathbb{1}\{|\eta(X) - \theta^*| \leqslant t\}\big] + \\
&\quad + 2\mathbb{E}[|\eta(X) - \hat{\eta}(X)| \Big(\tfrac{2|\eta(X) - \hat{\eta}(X)|}{t}\Big)^{q-1} \mathbb{1}\{|\eta(X) - \theta^*| > t\}] + \\
&\quad + 2|\hat{\theta} - \theta^*| \, \mathbb{P}\big(|\eta(X) - \theta^*| \leq 2|\hat{\theta} - \theta^*|\big) \\
&\leqslant 2M^{1-\frac{1}{q}} \|\eta - \hat{\eta}\|_q \, t^{\alpha(1-\frac{1}{q})} + 2^q \, \frac{\|\eta - \hat{\eta}\|_q^q}{t^{q-1}} + 2^{1+\alpha} M |\hat{\theta} - \theta^*|^{1+\alpha},
\end{aligned}$$

using $2|\eta(X) - \hat{\eta}(X)| \geq |\eta(X) - \theta^*| > t$ on that event, Hölder's inequality and Assumption 3.

Choosing $t = 2^{\frac{q}{q+\alpha}} \left(\frac{q}{\alpha}\right)^{\frac{q}{(q-1)(q+\alpha)}} M^{-\frac{1}{q+\alpha}} \|\eta - \hat{\eta}\|_q^{\frac{q}{q+\alpha}}$ (see Audibert & Tsybakov (2007)), we get

$$\frac{U(g_{\theta^*}) - U(\hat{g})}{c_1 K_0} \leq 2^{\frac{q(1+\alpha)}{q+\alpha}} \, \frac{q+\alpha}{q} \Big(\frac{q}{\alpha}\Big)^{\frac{\alpha}{q+\alpha}} M^{\frac{q-1}{q+\alpha}} \|\eta - \hat{\eta}\|_q^{\frac{q(1+\alpha)}{q+\alpha}} + 2^{1+\alpha} M |\hat{\theta} - \theta^*|^{1+\alpha}.$$

To control the last term notice that

$$|\hat{\theta} - \theta^*|^{1+\alpha} \leqslant 2^{\alpha} (|\hat{\theta} - \bar{\theta}|^{1+\alpha} + |\bar{\theta} - \theta^*|^{1+\alpha}).$$

As regards the first term, by using the bisection algorithm, we know $|\hat{\theta} - \bar{\theta}| \leqslant 2^{-K_{\max}}$ almost surely, where $K_{\max}$ is the number of iterations. Then we simply have:

$$|\hat{\theta} - \bar{\theta}|^{1+\alpha} \leqslant 2^{-(1+\alpha)K_{\max}} \quad \text{a.s.}$$

Let's focus now on the second term and, similarly to the proof of Proposition 1, we have:

$$|\bar{\theta} - \theta^*|^{1+\alpha} \leqslant K_1^{1+\alpha} \|\hat{F}_{\hat{\eta}} - F_\eta\|_1^{1+\alpha} \leqslant 2^{\alpha} K_1^{1+\alpha} \left(\|F_\eta - \hat{F}_\eta\|_1^{1+\alpha} + \|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\|_1^{1+\alpha}\right). \tag{21}$$

Combining with the previous display,

$$2^{1+\alpha} M |\hat{\theta} - \theta^*|^{1+\alpha} \leq 2^{1+2\alpha} M \, 2^{-(1+\alpha)K_{\max}} + 2^{1+3\alpha} M K_1^{1+\alpha} \left(\|F_\eta - \hat{F}_\eta\|_1^{1+\alpha} + \|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\|_1^{1+\alpha}\right).$$

For the first term we use:

$$\mathbb{E}_{\mathcal{D}_N^U}\|F_\eta - \hat{F}_\eta\|_1^{1+\alpha} = \mathbb{E}_{\mathcal{D}_N^U}\left(\int_0^1 |P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t)|dt\right)^{1+\alpha}$$
$$\leqslant \int_0^1 \mathbb{E}_{\mathcal{D}_N^U}|P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t)|^{1+\alpha}dt.$$

and, with $p(t) = P(\eta(X) \geqslant t)$, we get:

$$\begin{aligned}
&\mathbb{E}_{\mathcal{D}_N^U}|P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t)|^{1+\alpha} \\
&\quad= \int_0^{+\infty} P_{\mathcal{D}_N^U}\left(|P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t)|^{1+\alpha} \geqslant x\right)dx \\
&\quad= \int_0^{+\infty} P_{\mathcal{D}_N^U}\left(|P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t)| \geqslant x^{\frac{1}{1+\alpha}}\right)dx \\
&\quad= \int_0^{+\infty} P_{\mathcal{D}_N^U}\left(|P(\eta(X) \geqslant t) - \hat{P}_N(\eta(X) \geqslant t)| \geqslant u\right)(1+\alpha)u^\alpha du \\
&\quad\leqslant 2(1+\alpha)\int_0^{+\infty} \exp\left(-\frac{Nu^2}{2(p(t)+\frac{1}{3}u)}\right)u^\alpha du \\
&\quad= 2(1+\alpha)\left(\int_0^{3p(t)} u^\alpha \exp\left(-\frac{Nu^2}{4p(t)}\right)du + \int_{3p(t)}^{+\infty} u^\alpha \exp\left(-\frac{Nu}{4}\right)du\right) \\
&\quad\leqslant (1+\alpha)\left(\left(\frac{4p(t)}{N}\right)^{\frac{1+\alpha}{2}}\Gamma\left(\frac{1+\alpha}{2}\right) + 2\left(\frac{4}{N}\right)^{1+\alpha}\Gamma(1+\alpha)\right).
\end{aligned}$$

Integrating over $t \in [0,1]$, and using $\int_0^1 p(t)^{\frac{1+\alpha}{2}}dt \leq \int_0^1 p(t)^{\frac{1}{2}}dt \leq \sqrt{p}$ (since $p(t) \leq 1$, and by Cauchy–Schwarz with $\int_0^1 p(t)\,dt = \mathbb{E}[\eta(X)] = p$), we get

$$\mathbb{E}_{\mathcal{D}_N^U}\|F_\eta - \hat{F}_\eta\|_1^{1+\alpha} \leq (1+\alpha)\Big(2^{1+\alpha}\sqrt{p}\,\Gamma\big(\tfrac{1+\alpha}{2}\big)\frac{1}{N^{\frac{1+\alpha}{2}}} + 2^{3+2\alpha}\,\Gamma(1+\alpha)\frac{1}{N^{1+\alpha}}\Big).$$

Finally,

$$\left\|\hat{F}_{\hat{\eta}} - \hat{F}_\eta\right\|_1^{1+\alpha} = \left(\inf_{\omega\in\mathcal{S}_N}\frac{1}{N}\sum_{i=n+1}^{n+N}\left|Z_i - \hat{Z}_{\omega(i)}\right|\right)^{1+\alpha} \leqslant \left(\frac{\sum_{i=n+1}^{n+N}|\eta(X_i) - \hat{\eta}(X_i)|}{N}\right)^{1+\alpha}.$$

We use again that $(a+b)^{1+\alpha} \leqslant 2^\alpha a^{1+\alpha} + 2^\alpha b^{1+\alpha}$ and we get

$$\begin{aligned}
\left(\frac{\sum_{i=n+1}^{n+N}|\eta(X_i) - \hat{\eta}(X_i)|}{N}\right)^{1+\alpha} &\leqslant 2^\alpha\left(\frac{\sum_{i=n+1}^{n+N}|\eta(X_i) - \hat{\eta}(X_i)|}{N} - \|\eta(X) - \hat{\eta}(X)\|_1\right)^{1+\alpha} \\
&\quad + 2^\alpha\|\eta(X) - \hat{\eta}(X)\|_1^{1+\alpha}.
\end{aligned}$$

We can control the first term using Hoeffding's inequality:

$$\mathbb{P}\left(\left|\frac{1}{N}\sum_{i=n+1}^{n+N}|\eta(X_i) - \hat{\eta}(X_i)| - \|\eta(X) - \hat{\eta}(X)\|_1\right|^{1+\alpha} \geqslant \left(\frac{\log 2/\delta}{2N}\right)^{\frac{1+\alpha}{2}}\right) \leqslant \delta.$$

To obtain a bound in expectation we compute

$$\begin{aligned}
&\mathbb{E}_{\mathcal{D}_N^U}\left[\left|\frac{1}{N}\sum_{i=n+1}^{n+N}|\eta(X_i) - \hat{\eta}(X_i)| - \|\eta(X) - \hat{\eta}(X)\|_1\right|^{1+\alpha}\right] = \\
&\quad= \int_0^{+\infty}\mathbb{P}\left(\left|\frac{1}{N}\sum_{i=n+1}^{n+N}|\eta(X_i) - \hat{\eta}(X_i)| - \|\eta(X) - \hat{\eta}(X)\|_1\right|^{1+\alpha} > s\right)ds \leqslant \\
&\quad\leqslant \frac{1+\alpha}{(2N)^{\frac{1+\alpha}{2}}}\Gamma(\frac{1+\alpha}{2}).
\end{aligned}$$

Then

$$\mathbb{E}_{\mathcal{D}_N^U}\left\|\hat{F}_{\hat{\eta}}-\hat{F}_{\eta}\right\|_1^{1+\alpha}\leqslant(1+\alpha)2^{\frac{\alpha-1}{2}}\frac{1}{N^{\frac{1+\alpha}{2}}}\Gamma(\frac{1+\alpha}{2})+2^{\alpha}\|\eta(X)-\hat{\eta}(X)\|_1^{1+\alpha}.$$

Putting things together we recover the final bound:

$$\begin{aligned}\frac{U(g_{\theta^*})-\mathbb{E}_{\mathcal{D}_N^U}U(\hat{g})}{c_1K_0}\leqslant&2^{\frac{q(1+\alpha)}{q+\alpha}}\frac{q+\alpha}{q}\Big(\frac{q}{\alpha}\Big)^{\frac{\alpha}{q+\alpha}}M^{\frac{q-1}{q+\alpha}}\|\eta-\hat{\eta}\|_q^{\frac{q(1+\alpha)}{q+\alpha}}+ \qquad (22)\\ &+(1+\alpha)\Gamma\Big(\tfrac{1+\alpha}{2}\Big)MK_1^{1+\alpha}\Big(2^{2+4\alpha}\sqrt{p}+2^{\frac{1+7\alpha}{2}}\Big)\frac{1}{N^{\frac{1+\alpha}{2}}}+\\ &+2^{4+5\alpha}(1+\alpha)\Gamma(1+\alpha)MK_1^{1+\alpha}\frac{1}{N^{1+\alpha}}+\\ &+2^{1+4\alpha}MK_1^{1+\alpha}\|\eta-\hat{\eta}\|_1^{1+\alpha}+\\ &+2^{1+2\alpha}M\,2^{-(1+\alpha)K_{\max}}. \qquad (23)\end{aligned}$$

Finally, taking the expectation $\mathbb{E}_{\mathcal{D}_n^L}$ over the labelled dataset we obtain the results. □

B Experimental Details

Datasets. We evaluate on different binary classification benchmarks covering a wide range of sample sizes, dimensionalities, and class priors, including highly imbalanced problems ($p < 0.1$). *german.numer* ($n = 1000$, $d = 24$, $p = 0.300$) is the LIBSVM encoding of the UCI statlog German credit dataset, where the positive class corresponds to bad-credit applicants. *Spambase* ($n = 4601$, $d = 57$, $p = 0.394$) classifies e-mails as spam or legitimate based on word and character frequencies. *Sonar* ($n = 208$, $d = 60$, $p = 0.534$) distinguishes sonar returns bouncing off metal cylinders (mines) from those bouncing off rocks, using 60 energy measurements. *Phoneme* ($n = 5404$, $d = 5$, $p = 0.294$) distinguishes nasal from oral phonemes using five spectral features. *Adult* ($n \approx 45{,}222$ after removing missing values, $d = 14$, $p = 0.239$) asks whether a person's annual income exceeds \$50K from census attributes. The last two datasets are highly imbalanced. *Creditcard* ($n = 284{,}807$, $d = 30$, $p = 0.0017$) contains credit card transactions made by European cardholders, described by 28 PCA-transformed components plus the time and amount of the transaction; the positive class (fraud) accounts for only 492 transactions. *Mammography* ($n = 11{,}183$, $d = 6$, $p = 0.0232$) is a medical benchmark for the detection of microcalcifications in digitized mammograms from six image-derived features, with 260 positive cases.

Training protocol. Each dataset is split into 80% pool and 20% held-out test set over 50 independent random trials. The pool is divided equally into a regression half (used to fit $\hat{\eta}$) and a threshold half (used to select $\hat{\theta}$). All features are normalised with min–max scaling fitted exclusively on the regression half. We compare three regression functions: ordinary least-squares linear regression (LinReg), ℓ_2-regularised logistic regression (LR), and kernel ridge regression with a Gaussian (RBF) kernel (KRR). For LR the regularisation parameter C is chosen from $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ by 3-fold cross-validation on the regression half maximising F_1. For KRR the bandwidth $\gamma \in \{10^{-3}, 5{\times}10^{-3}, 0.02, 0.05, 0.1, 0.5\}$ and regularization $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1\}$ are jointly selected by the same 3-fold procedure. On the Adult dataset, KRR training and its cross-validation are restricted to a random subsample of 2000 points from the regression half to keep the kernel matrix computationally feasible. Hyperparameter selection is performed before the threshold-estimation step and is excluded from the reported running times. Three threshold strategies are compared: (i) Default, which fixes $\hat{\theta} = 0.5$; (ii) 10-fold CV, which selects $\hat{\theta}$ by maximising F_1 on the threshold half via 10-fold cross-validation over a fixed uniform grid of 40 equally spaced values in $[10^{-3}, 1 - 10^{-3}]$ (uses labels); and (iii) Bisection (our method), which solves the empirical fixed-point equation on the feature vectors of the threshold half alone, requiring no labels beyond the class prior $\hat{p}$ estimated from the regression half.

NeurIPS Paper Checklist

1. **Claims**

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

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